

REPORT
ON
MINI PROJECT

SUBMITTED BY
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Salary Survey Insights: Analytics In Action

Domain: Human Resources Analytics

Tools Used:

- Excel
- Power BI

Submission Date: 21- Feb – 2026

Mentor Name: Kumaran M

Raw Dataset Link: [Salary Survey – Raw Data](#)

Cleaned Dataset Link: [Salary Survey – Cleaned Data](#)

Salary Survey Insights: Analytics in Action

Objective:

This mini project aims to analyze and visualize salary survey data collected from online participants. The goal is to uncover compensation trends across demographics, qualifications, industries, and work locations, while ensuring the dataset is cleaned and transformed for accurate insights.

Dataset Overview:

The dataset includes participant-entered information such as:

- Age
- Qualification
- Total Experience & Relevant Experience
- Industry
- Work Location
- Currency Type
- Race & Gender
- Annual Salary & Incentives

Data Cleaning & Transformation (Excel)

- Handled missing and inconsistent values using XLOOKUP, VLOOKUP, INDEX, and MATCH functions.
- Standardized currency types and work location entries for consistency.
- Added derived columns for better segmentation, such as:
- Experience Category (Beginner, Early Career, Intermediate, Experienced, Advanced Professional, Senior Professional, Veteran, Expert)
- Age Category (Junior, Youth, Young-Adult, Adult, Mid-Senior, Senior, Elder)

- Calculated Total Salary by combining Annual Salary and Incentives into a single column for accurate compensation analysis.

Visualization & Analysis (Power BI)

- Created measures for average salary, incentive percentage, and salary distribution by demographics.
- Designed interactive dashboards with charts such as:
- Salary by Industry & Work Location
- Salary vs. Experience Level
- Gender & Race-based compensation comparisons
- Incentives contribution to total compensation
- Added slicers for dynamic filtering (e.g., by qualification, industry, or location).

Insights & Findings:

The analysis helps answer key questions like:

- How does experience impact annual salary across industries?
- Are there noticeable gender or race-based disparities in compensation?
- Which industries and locations offer the highest salaries?
- What is the relationship between incentives and base salary?
- How do qualifications influence earning potential?

Data Cleaning:

Handling Missing Values:

In the Qualification field, blank entries were standardized and replaced with the value 'Unknown' to ensure consistency during analysis.

fx =IF(P2="", "Unknown", P2)				
O	P	Q	R	S
levant Experience	Qualification	Gender	Race	Education
7 years	Master's degree	Woman	White	Master's degree
7 years	College degree	Non-binary	White	College degree
4 years	College degree	Woman	White	College degree
7 years	College degree	Woman	White	College degree
7 years	College degree	Woman	White	College degree
4 years	Master's degree	Man	White	Master's degree
4 years	College degree	Woman	White	College degree
7 years	Master's degree	Man	White	Master's degree
- 30 years	College degree	Woman	White	College degree
- 30 years	College degree	Woman	Hispanic, Latino, or Spanish origin, White	College degree
7 years	College degree	Woman	White	College degree
7 years	PhD	Woman	Hispanic, Latino, or Spanish origin, White	PhD
- 20 years	College degree	Man	Asian or Asian American, White	College degree
4 years		Woman	White	Unknown
ear or less	College degree	Woman	White	College degree
7 years	College degree	Man	White	College degree
10 years	Some college	Woman	White	Some college
- 30 years	College degree	Woman	White	College degree
4 years	Master's degree	Woman	White	Master's degree
- 20 years	Master's degree	Woman	White	Master's degree
7 years	Master's degree	Woman	White	Master's degree
4 years	PhD	Woman	White	PhD
10 years	Master's degree	Woman	White	Master's degree
4 years	PhD	Woman	Asian or Asian American	PhD
4 years		Woman	White	Unknown
ear or less	Master's degree	Woman	White	Master's degree
4 years	College degree	Man	White	College degree

Gender field left blank has been assigned 'Unknown' for consistency.

fx =IF([@Gender]="", "Unknown", [@Gender])		
	W	X
1	Gender	Gender1
e	Woman	Woman
	Man	Man
e	Man	Man
æe		Unknown
æe	Woman	Woman
æe	Woman	Woman
	Other or prefer not to ans	Other or prefer not to ans
æe	Woman	Woman
e	Man	Man
	Woman	Woman
e	Woman	Woman
	Man	Man
æe	Woman	Woman
e	Woman	Woman
e	Woman	Woman

Standardizing Data Types:

The Incentives column was converted and standardized to a numeric data type to ensure accurate aggregation and reliable salary-related calculations.

Microsoft Excel ribbon showing the 'Number' format selected for the 'Incentives' column.

Formula bar: `=IF(TRIM(G2)="",0,G2)`

Industry	Job title	Additional Job Details	Annual Salary	Incentives	Incentive Amount
Education	High Research and Instruction Librarian		55,000	0.00	0
Computing or Technology	Change & Internal Communications Manager		54,600	0.00	0
Accounting	Bar Marketing Specialist		34,000		0
Nonprofits	Program Manager		62,000	3000	3,000
Accounting	Bar Accounting Manager		60,000	7000	7,000
Education	High Scholarly Publishing Librarian		62,000		0
Publishing	Publishing Assistant		33,000	2000	2,000
Education	Prim Librarian	High school, FT	50,000		0
Computing or Technology	Systems Analyst	Data developer/ETL Developer	112,000	10000	10,000
Accounting	Bar Senior Accountant		45,000	0	0
Nonprofits	Office Manager		47,500	0	0
Education	High Deputy Title IX Coordinator/ Assistant Director Office of Equity and		62,000	0	0
Accounting	Bar Manager of Information Services		100,000	0	0
Law	Legal Aid Staff	non-profit law firm	52,000	0	0
Health care	Patient care coordinator		32,000		0
Utilities & Telecommunications	Quality And Compliance Specialist		24,000	500	500
Business or Communications	Executive Assistant		85,000	5000	5,000
Art & Design	graphic designer		59,000		0
Business or Communications	Senior Manager		98,000	1000	1,000
Education	High Assistant Director of Academic Advising		54,000		0
Health care	Data Programmer Analyst		74,000	0	0
Nonprofits	Program Coordinator & Assistant Editor		50,000		0
Nonprofits	Event Planner		63,000		0
Government and Public Administration	Researcher		96,000	1000	1,000
Public Library	Teen Librarian		44,500	0	0
Education	High Communications Specialist		60,000	0	0
Nonprofits	Program Director	I manage staff members who each manage 8-15 other staff	62,000		0
Law	Bookkeeper/Billing Manager		48,000	4000	4,000
Government and Public Administration	Economist		140,000		0
Engineering or Technology	Research Engineer		80,000		0
Nonprofits	Volunteer and Giving Coordinator		39,000		0
Media & Digital	Editor		125,000	0	0
Accounting	Bar Financial Advisor		230,000		0
Accounting	Bar Accounting Manager		110,000	15000	15,000

Derived Columns:

From the datetime column, the date, time, and year components were extracted into separate fields to simplify analysis and enable clearer insights.

✕ ✓ fx		=YEAR([@Date])				
AC	AD	AE	AF	AG	AH	AI
	Time	Total Sale	Year			
4/27/2021	11:02:10 AM	55,000	2021			
4/27/2021	11:02:22 AM	54,600	2021			
4/27/2021	11:02:38 AM	34,000	2021			
4/27/2021	11:02:41 AM	65,000	2021			
4/27/2021	11:02:42 AM	67,000	2021			
4/27/2021	11:02:46 AM	62,000	2021			
4/27/2021	11:02:51 AM	35,000	2021			
4/27/2021	11:03:00 AM	50,000	2021			
4/27/2021	11:03:01 AM	122,000	2021			
4/27/2021	11:03:02 AM	45,000	2021			
4/27/2021	11:03:03 AM	47,500	2021			
4/27/2021	11:03:07 AM	62,000	2021			
4/27/2021	11:03:09 AM	100,000	2021			
4/27/2021	11:03:10 AM	52,000	2021			
4/27/2021	11:03:11 AM	32,000	2021			
4/27/2021	11:03:11 AM	24,500	2021			
4/27/2021	11:03:13 AM	90,000	2021			
4/27/2021	11:03:19 AM	59,000	2021			
4/27/2021	11:03:21 AM	99,000	2021			
4/27/2021	11:03:23 AM	54,000	2021			
4/27/2021	11:03:24 AM	74,000	2021			
4/27/2021	11:03:24 AM	50,000	2021			
4/27/2021	11:03:27 AM	63,000	2021			
4/27/2021	11:03:29 AM	97,000	2021			

Sort Smallest to Largest

Sort Largest to Smallest

Sort by Color

Sheet View

Clear Filter From "Year"

Filter by Color

Number Filters

Search

(Select All)

2021

2022

2023

2024

2025

2026

</

A new column Total Salary was created by combining Annual Salary and Incentives to provide a complete view of participant compensation.

Font		Alignment		Number	
Arial 10 A [^] A ^v		Wrap Text		Number	
B I U		Merge & Center		\$ %	
f _x		=[@[Annual Salary]]+[@[Incentive Amount]]			
V	W	X	Y	Z	
	Total Salary				
ee	55,000				
e	54,600				
e	34,000				
e	65,000				
e	67,000				
ee	62,000				
e	35,000				
ee	50,000				
e	122,000				
e	45,000				
e	47,500				
	62,000				
e	100,000				
	52,000				
e	32,000				
e	24,500				
	90,000				
e	59,000				
ee	99,000				
ee	54,000				
ee	74,000				
	50,000				
ee	63,000				
	97,000				
	44,500				
ee	60,000				
e	62,000				
e	52,000				
ee	140,000				
e	80,000				
e	39,000				
ee	125,000				
e	230,000				
e	125,000				

Working Country Data Cleaning & Standardization:

Step 1 → Added ISO Country List

A master sheet (CountryList) was created with ISO-standard country names to serve as the reference table.

Name
Afghanistan
Aland Islands
Albania
Algeria
American Samoa
Andorra
Angola
Anguilla
Antarctica
Antigua and Barbuda
Argentina
Armenia
Aruba
Australia
Austria
Azerbaijan
Bahamas
Bahrain
Bangladesh
Barbados
Belarus
Belgium
Belize

Step 2 → Removed Country Names from Sentences

Country names embedded in text fields were extracted and removed using Excel formulas.

M2

Work Country	Country
Aotearoa New Zealand	New Zealand
ARGENTINA BUT MY ORG IS IN THAILAND	Argentina
Austria, but I work remotely for a Dutch/British company	Austria
bonus based on meeting yearly goals set w/ my supervisor	US
Bosnia and Herzegovina	Bosnia and Herzegovina
Canada and USA	Canada
Canada, Ottawa, ontario	Canada
Česká republika	Česká republika
Company in Germany. I work from Pakistan.	Germany
England, Gb	England, Gb
England, United Kingdom	United Kingdom
England, United Kingdom	United Kingdom
England, United Kingdom	United Kingdom
England/UK	UK
For the United States government, but posted overseas	United States
From New Zealand but on projects across APAC	New Zealand
From Romania, but for an US based company	Oman
I am located in Canada but I work for a company in the US	Canada
I earn commission on sales. If I meet quota. I'm guaranteed another 16k mi I earn commission on sales. If I meet quota. I'm guaranteed another 16k	

Step 3 → Generated Unique Country List

A distinct list of countries was compiled in another sheet to identify variations and inconsistencies.

\$2,175.84/year is deducted for ben

1
Afghanistan
Africa
America
Argentina
Australia
Austria
Bahamas
Bangladesh
Belgium
Bermuda
Bosnia and Herzegovina
Brazil
Brazil
Britain
Bulgaria
Burma
California
Cambodia
Can
Canada
Canada
Canadá
Canadw
Canda
Catalonia
Cayman Islands
Česká republika
Chile
China
Colombia
Congo
Contracts
Costa Rica

Step 4 → Created Error vs. Correct Mapping Table

Incorrect spellings were mapped against corrected values in a dedicated table.

ErrorValue	CorrectValue
Austria	Australia
Scotland	United Kingdom
England	United Kingdom
Brasil	Brazil
California	United States
Can	Canada
Canad	Canada
Canadá	Canada
Canadw	Canada
Canda	Canada
hong konh	Hong Kong
Italia	Italy
London	United Kingdom
Nederland	Netherlands
U. S	United States
U. S.	United States
U. S.	United States
U. S.	United States
U.A.	United States
U.K	United Kingdom
U.K.	United Kingdom
U.K.	United Kingdom
U.K. (northern Eng	United Kingdom
U.S	United States
U.S.	United States
U.S.	United States
U.S.	United States
U.S.A	United States
U.S.A	United States
U.S.A.	United States
U.S.A.	United States
U.S>	United States
U.SA	United States
UA	United States
UAE	United Arab Emire

Step 5 → Corrected Country Names in Main Survey Sheet

Using the mapping table, incorrect spellings were replaced with standardized names.

fx =IFERROR(VLOOKUP(M2,CountryList!\$E\$1:\$F\$76,2,FALSE),M2)	
Country1	Country2
United States	United States
United Kingdom	United Kingdom
US	United States
USA	United States
US	United States
USA	United States
USA	United States
United States	United States
US	United States

Step 6 → Validated with VLOOKUP

Country names were checked against the CountryList. If not found, the value was set to “Unknown”.

fx	=IFERROR(VLOOKUP(UPPER(TRIM(O2)),CountryList!\$A:\$A,1,FALSE),"Unknown")
----	--

Step 7 → Final Standardization

All country names were standardized, ensuring consistency across the dataset for reliable analysis.

Country_after_correction	Country
Belgium	Belgium
\$2,175.84/year is deducted for benefits	Unknown
Afghanistan	Afghanistan
Belgium	Belgium
Belgium	Belgium
Bangladesh	Bangladesh
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Argentina	Argentina
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Belgium	Belgium
Argentina	Argentina
Belgium	Belgium

Age Category Mapping

To ensure consistency in reporting, an AgeCategory table was created to classify individuals into predefined age ranges. The mapping of age values to their respective categories was performed using the VLOOKUP function in Excel. This approach standardized age data and facilitated clear segmentation for analysis.

fx =VLOOKUP(B2,AgeCategory[#All],2,FALSE)

	B	C	D
	Age Range	Age Category	Industry
	25-34	Young-Adult	Education (Primary
	45-54	Mid-Senior	Computing or Tech
	35-44	Adult	Accounting, Bankin
	25-34	Young-Adult	Nonprofits
	35-44	Adult	Education (Higher E
	35-44	Adult	Accounting, Bankin
	25-34	Young-Adult	Law
	18-24	Youth	Health care
	35-44	Adult	Utilities & Telecomr
	35-44	Adult	Business or Consu
	45-54	Mid-Senior	Art & Design
	35-44	Adult	Business or Consu
	35-44	Adult	Education (Higher E
	25-34	Young-Adult	Health care
	35-44	Adult	Nonprofits
	35-44	Adult	Nonprofits
	35-44	Adult	Government and Pu
	25-34	Young-Adult	Public Library
	35-44	Adult	Education (Higher E
	25-34	Young-Adult	Nonprofits
	35-44	Adult	Law

Age Range	Age Category
under 18	Junior
18-24	Youth
25-34	Young-Adult
35-44	Adult
45-54	Mid-Senior
55-64	Senior
65 or over	Elder

Experience Category Mapping:

Created ExpCategory table to define experience levels.

Experience Range	Category	Level
1 year or less	Beginner	1
2 - 4 years	Early Career	2
5-7 years	Intermediate	3
8 - 10 years	Experienced	4
11 - 20 years	Advanced Professional	5
21 - 30 years	Senior Professional	6
31 - 40 years	Veteran	7
41 years or more	Expert	8

Added new column Experience Category based on total experience range. Each employee's total experience mapped to the corresponding category.

fx =VLOOKUP([@[Total Experience]],ExpCategory[#All],2,FALSE)	
T	U
Total Experience	Exp Level
5-7 years	Intermediate
8 - 10 years	Experienced
2 - 4 years	Early Career
8 - 10 years	Experienced
8 - 10 years	Experienced
8 - 10 years	Experienced
2 - 4 years	Early Career
5-7 years	Intermediate
21 - 30 years	Senior Professional
21 - 30 years	Senior Professional
5-7 years	Intermediate
11 - 20 years	Advanced Professional
11 - 20 years	Advanced Professional
2 - 4 years	Early Career
1 year or less	Beginner
11 - 20 years	Advanced Professional
8 - 10 years	Experienced
21 - 30 years	Senior Professional
11 - 20 years	Advanced Professional
11 - 20 years	Advanced Professional
5-7 years	Intermediate
5-7 years	Intermediate
11 - 20 years	Advanced Professional
8 - 10 years	Experienced

Handling Experience Inconsistency:

Created Experience Status column to identify inconsistency when Total Experience is less than Relevant Experience.

f_x =VLOOKUP([@[Total Experience]],ExpCategory[#All],3,FALSE) >= VLOOKUP([@[Relevant Experience]],ExpCategory[#All],3,FALSE)

T	U	V	W	X	Y	
Total Experience	Exp Level	Relevant Experience	Exp Status	Qualificatio	Qualification1	Gender
5-7 years	Intermediate	5-7 years	TRUE	Master's degree	Master's degree	Woman
8 - 10 years	Experienced	5-7 years	TRUE	College degree	College degree	Non-t
2 - 4 years	Early Career	2 - 4 years	TRUE	College degree	College degree	Woman
8 - 10 years	Experienced	5-7 years	TRUE	College degree	College degree	Woman
8 - 10 years	Experienced	5-7 years	TRUE	College degree	College degree	Woman
8 - 10 years	Experienced	2 - 4 years	TRUE	Master's degree	Master's degree	Man
2 - 4 years	Early Career	2 - 4 years	TRUE	College degree	College degree	Woman
5-7 years	Intermediate	5-7 years	TRUE	Master's degree	Master's degree	Man
21 - 30 years	Senior Profession	21 - 30 years	TRUE	College degree	College degree	Woman
21 - 30 years	Senior Profession	21 - 30 years	TRUE	College degree	College degree	Woman
5-7 years	Intermediate	5-7 years	TRUE	College degree	College degree	Woman
11 - 20 years	Advanced Profess	5-7 years	TRUE	PhD	PhD	Woman
11 - 20 years	Advanced Profess	11 - 20 years	TRUE	College degree	College degree	Man
2 - 4 years	Early Career	2 - 4 years	TRUE		Unknown	Woman
1 year or less	Beginner	1 year or less	TRUE	College degree	College degree	Woman
11 - 20 years	Advanced Profess	5-7 years	TRUE	College degree	College degree	Man
8 - 10 years	Experienced	8 - 10 years	TRUE	Some college	Some college	Woman
21 - 30 years	Senior Profession	21 - 30 years	TRUE	College degree	College degree	Woman
11 - 20 years	Advanced Profess	2 - 4 years	TRUE	Master's degree	Master's degree	Woman

For records with Exp Status = FALSE, values of Total Experience and Relevant Experience were interchanged to correct inconsistency.

f_x =IF([@[Exp Status]]=FALSE,[@[Relevant Experience]],[@[Total Experience]])

U	V	W	X	Y	Z
Total Experience	Exp Level	Relevant Experience	Exp Status	Tot Exp New	Relv Exp New
5-7 years	Intermediate	5-7 years	TRUE	5-7 years	5-7 years
5-7 years	Intermediate	8 - 10 years	FALSE	8 - 10 years	5-7 years
8 - 10 years	Experienced	8 - 10 years	TRUE	8 - 10 years	8 - 10 years
11 - 20 years	Advanced Profess	8 - 10 years	TRUE	11 - 20 years	8 - 10 years
5-7 years	Intermediate	5-7 years	TRUE	5-7 years	5-7 years
31 - 40 years	Veteran	11 - 20 years	TRUE	31 - 40 years	11 - 20 years
8 - 10 years	Experienced	5-7 years	TRUE	8 - 10 years	5-7 years
8 - 10 years	Experienced	8 - 10 years	TRUE	8 - 10 years	8 - 10 years
5-7 years	Intermediate	2 - 4 years	TRUE	5-7 years	2 - 4 years

Currency Mapping:

Created CurrencyTable with Country Name and Currency Code.

A	B	C
Country	Currency	Code
Afghanistan	Afghani	AFN
Albania	Lek	ALL
Algeria	Algerian Dinar	DZD
American Samoa	US Dollar	USD
Angola	Kwanza	AOA
Anguilla	East Caribbean Dollar	XCD
Antigua and Barbuda	East Caribbean Dollar	XCD
Argentina	Argentine Peso	ARS
Armenia	Armenian Dram	AMD
Aruba	Aruban Florin	AWG
Australia	Australian Dollar	AUD
Austria	Austrian Shilling*	ATS
Azerbaijan	Azerbaijan Manat	AZN
Bahamas	Bahamian Dollar	BSD
Bahrain	Bahraini Dollar	BHD
Bangladesh	Taka	BDT
Barbados	Barbados Dollar	BBD
Belarus	Belarusian Rubel	BYN
Belgium	Belgian Franc*	BEF
Belize	Belize Dollar	BZD
Benin	CFA Franc	XOF
Bermuda	Bermudian Dollar	BMD
Bhutan	Ngultrum	BTN

For records where currency value was marked as Other, the correct currency was set by checking the corresponding Country Name.

fx =IF([@Currency]="Other",XLOOKUP([@Country],CurrencyTable[Country],CurrencyTable[Code],"Not Found"),[@Currency])						
J	K	L	M	N	O	P
Currency	Other Currency	CurrencyType	Extra Income	Work Country	Country1	Country_After Trim
Other	THAI BAHT	THB		THAILAND	Thailand	Thailand
Other	INR	INR		India	India	India
Other	Peso Argentino	ARS	Exchange ab	Argentina	Argentina	Argentina
Other	MYR	MYR		Malaysia	Malaysia	Malaysia
Other	Czk	CZK		Česká republika	Česká republika	Česká republika
SEK		SEK		Sweden	Sweden	Sweden
Other	CHF	CHF	While I work	Switzerland	Switzerland	Switzerland
Other	NOK	NOK		Norway	Norway	Norway
Other	NOK	NOK		Norway	Norway	Norway
Other	USD	USD		USA	USA	USA
Other	BR\$	BRL	For reference	Brazil	Brazil	Brazil
SEK		SEK		Sweden	Sweden	Sweden
SEK		SEK		Sweden	Sweden	Sweden
Other	SEK	SEK		Sweden	Sweden	Sweden
Other	Dkk	DKK		Denmark	Denmark	Denmark
Other	EUR	HUF		Hungary	Hungary	Hungary
SEK		SEK		Sweden	Sweden	Sweden
SEK		SEK		US	US	US
Other	NOK	NOK		Norway	Norway	Norway
Other	TTD	TTD		Trinidad and Tobag	Trinidad and Toba	Trinidad and Toba
Other	Indian rupees	INR		India	India	India
Other	BRL (R\$)	BRL		Brazil	Brazil	Brazil
Other	Mexican pesos	MXN		Mexico	Mexico	Mexico
Other	CZK	CZK		Czech republic	Czech Republic	Czech Republic
Other	CZK	CZK		Czechia	Czechia	Czechia
Other	GBP	GBP		Scotland	Scotland	Scotland
Other	NOK	NOK		Norway	Norway	Norway

Industry Classification:

Created IndustryTable in the Industry sheet.

Industry Type
Accounting, Banking & Finance
Agriculture or Forestry
Art & Design
Business or Consulting
Computing or Tech
Education (Primary/Secondary)
Education (Higher Education)
Engineering or Manufacturing
Entertainment
Government and Public Administration
Health care
Hospitality & Events
Insurance
Law
Law Enforcement & Security
Leisure, Sport & Tourism
Marketing, Advertising & PR
Media & Digital
Nonprofits
Property or Construction
Recruitment or HR
Retail
Sales
Social Work
Transport or Logistics
Utilities & Telecommunications

Checked Industry values; those not found were set as "Other"

f_x =IFERROR(VLOOKUP([@Industry],IndustryTable[Industry Type],1,FALSE),"Other")

Industry	Industry Other	Job title
Education (Higher Education)	Education (Higher Education)	Research
Computing or Tech	Computing or Tech	Change
Accounting, Banking & Finance	Accounting, Banking & Finance	Marketin
Nonprofits	Nonprofits	Program
Accounting, Banking & Finance	Accounting, Banking & Finance	Account
Education (Higher Education)	Education (Higher Education)	Scholar
Publishing	Other	Publishi
Education (Primary/Secondary)	Education (Primary/Secondary)	Libraria
Computing or Tech	Computing or Tech	Systems
Accounting, Banking & Finance	Accounting, Banking & Finance	Senior A
Nonprofits	Nonprofits	Office M
Education (Higher Education)	Education (Higher Education)	Deputy
Accounting, Banking & Finance	Accounting, Banking & Finance	Manage
Law	Law	Legal Ai
Health care	Health care	Patient c
Utilities & Telecommunications	Utilities & Telecommunications	Quality /
Business or Consulting	Business or Consulting	Executiv
Art & Design	Art & Design	graphic
Business or Consulting	Business or Consulting	Senior M
Education (Higher Education)	Education (Higher Education)	Assistar
Health care	Health care	Data Pr
Nonprofits	Nonprofits	Program
Nonprofits	Nonprofits	Event Pl
Government and Public Administration	Government and Public Administration	Researc
Public Library	Other	Teen Lit
Education (Higher Education)	Education (Higher Education)	Commui

Created IndustryTypeTable with IndustryType and associated keywords.

Industry Type	Keywords
Agriculture or Forestry	Horticulture
Agriculture or Forestry	Agriculture
Agriculture or Forestry	Landscape
Accounting, Banking & Finance	Administrative
Accounting, Banking & Finance	Fundraising
Accounting, Banking & Finance	Financial
Accounting, Banking & Finance	Accountant
Accounting, Banking & Finance	Payment
Accounting, Banking & Finance	Insurance
Accounting, Banking & Finance	Finance
Accounting, Banking & Finance	Gambling
Accounting, Banking & Finance	Association
Accounting, Banking & Finance	Economics
Accounting, Banking & Finance	Analyst
Business or Consulting	Beauty
Business or Consulting	Business
Business or Consulting	BPO
Business or Consulting	Call Center
Business or Consulting	Pest Control
Business or Consulting	Publishing

For records marked as "Other", Industry type was updated by matching keywords from IndustryTypeTable.

✓
fx

=LET(
txt, [@Industry],
found, ISNUMBER(SEARCH(IndustryTypeTable[Keywords], txt)),
matchIndex, MATCH(TRUE, found, 0),
IFERROR(IF([@[Industry Other]]="Other", INDEX(IndustryTypeTable[Industry Type], matchIndex), [@Industry]), "Unknown")
)

Category	Industry	Industry Other	Industry1	Job title
-Adult	Media & Digital	Media & Digital	Media & Digital	Product
-Adult	Computing or Tech	Computing Or Tech	Computing or Tech	Sales
-Adult	Health care	Health Care	Health care	Account
-Adult	Engineering or Manufacturing	Engineering Or Manufacturing	Engineering or Manufacturing	Engine
-Adult	Government and Public Administration	Government And Public Administration	Government and Public Administration	Assiste
-Adult	Other	Other	Unknown	Propos
-Adult	Health care	Health Care	Health care	Pharm
-Adult	Computing or Tech	Computing Or Tech	Computing or Tech	Techni
-Adult	Accounting, Banking & Finance	Accounting, Banking & Finance	Accounting, Banking & Finance	IT Dire
-Adult	Sales	Sales	Sales	Admini

Records with over-salary values and zero-salary values were excluded.

fx =OR(AND(AJ28175>10, AJ28175<10000000), N28175<>"USD")

AJ	AK	AL	AM
Total Salary	Year	OverSalary	
10,920	2025	TRUE	
0	2025	FALSE	
200,000	2025	TRUE	
0	2025	FALSE	
2	2025	TRUE	
41,600	2025	TRUE	
40,000	2025	TRUE	
127,000	2025	TRUE	
600,000	2025	TRUE	
63,492	2025	TRUE	
54,000	2025	TRUE	
72,020	2025	TRUE	
0	2025	TRUE	
19,000	2025	TRUE	
151,000	2026	TRUE	
130,000	2026	TRUE	
125,000	2026	TRUE	
140,000,000	2026	TRUE	
0	2026	FALSE	

Sample Cleaned Dataset:

Attached image shows the final cleaned dataset after applying all data preparation steps.

A	B	C	D	E	F	G	H	I	J
Date Time	Date	Time	Age Range	Age Category	Industry	Job title	Annual Salary	Incentives	Total Salary
4/27/2021 11:02:10	4/27/2021	11:02:10 AM	25-34	Young-Adult	Education (Higher Education)	Research and Instructor	55,000	0	55,000
4/27/2021 11:02:22	4/27/2021	11:02:22 AM	25-34	Young-Adult	Computing or Tech	Change & Internal Commr	54,600	0	54,600
4/27/2021 11:02:38	4/27/2021	11:02:38 AM	25-34	Young-Adult	Accounting, Banking & Finance	Marketing Specialist	34,000	0	34,000
4/27/2021 11:02:41	4/27/2021	11:02:41 AM	25-34	Young-Adult	Nonprofits	Program Manager	62,000	3,000	65,000
4/27/2021 11:02:42	4/27/2021	11:02:42 AM	25-34	Young-Adult	Accounting, Banking & Finance	Accounting Manager	60,000	7,000	67,000
4/27/2021 11:02:46	4/27/2021	11:02:46 AM	25-34	Young-Adult	Education (Higher Education)	Scholarly Publishing Libra	62,000	0	62,000
4/27/2021 11:02:51	4/27/2021	11:02:51 AM	25-34	Young-Adult	Business or Consulting	Publishing Assistant	33,000	2,000	35,000
4/27/2021 11:03:00	4/27/2021	11:03:00 AM	25-34	Young-Adult	Education (Primary/Secondary)	Librarian	50,000	0	50,000
4/27/2021 11:03:01	4/27/2021	11:03:01 AM	45-54	Mid-Senior	Computing or Tech	Systems Analyst	112,000	10,000	122,000

K	L	M	N	O	P	Q	R	S	T
CurrencyType	Country	US State	City	Total Experience	Relevant Experience	Experience Level	Qualification	Gender	Race
USD	United States	Massachusetts	Boston	5-7 years	5-7 years	Intermediate	Master's degree	Woman	White
GBP	United Kingdom		Cambridge	8 - 10 years	5-7 years	Experienced	College degree	Non-binary	White
USD	United States	Tennessee	Chattanooga	2 - 4 years	2 - 4 years	Early Career	College degree	Woman	White
USD	United States	Wisconsin	Milwaukee	8 - 10 years	5-7 years	Experienced	College degree	Woman	White
USD	United States	South Carolina	Greenville	8 - 10 years	5-7 years	Experienced	College degree	Woman	White
USD	United States	New Hampshire	Hanover	8 - 10 years	2 - 4 years	Experienced	Master's degree	Man	White
USD	United States	South Carolina	Columbia	2 - 4 years	2 - 4 years	Early Career	College degree	Woman	White
USD	United States	Arizona	Yuma	5-7 years	5-7 years	Intermediate	Master's degree	Man	White
USD	United States	Missouri	St. Louis	21 - 30 years	21 - 30 years	Senior Professional	College degree	Woman	White
USD	United States	Florida	Palm Coast	21 - 30 years	21 - 30 years	Senior Professional	College degree	Woman	Hispanic

Pivot Tables:

Key Insights:

Annual Participant Gender Breakdown:

- Women consistently form the largest group across all years, with a grand total of 21,417 out of 28,161 records.
- Men represent the second-largest category, totaling 5,522 across all years.
- Non-binary and Other/Prefer not to answer categories remain relatively small but present, showing inclusivity in reporting.
- Data from 2022–2026 is comparatively sparse, suggesting either incomplete collection or smaller sample sizes.

Annual Gender Breakdown							
	Man	Non-binary	Other or prefer not to answer	Prefer not to answer	Unknown	Woman	Grand Total
2021	5315	738	282	1	164	21086	27586
2022	97	4	8		1	213	323
2023	35	1	3		2	37	78
2024	46	3	3		4	43	99
2025	27				8	35	70
2026	2					3	5
Grand Total	5522	746	296	1	179	21417	28161

Industry Gender Breakdown:

- Women dominate across most industries, with a grand total of 21,417 compared to 5,522 men, showing a strong female representation overall.
- Higher Education stands out with a large female workforce (2,074) and notable representation of non-binary individuals (271).
- Nonprofits show a strong female majority (2,094) but also relatively higher non-binary representation (86) compared to other industries.
- Engineering/Manufacturing has a considerable male presence (503) but women still form the majority (1,318).

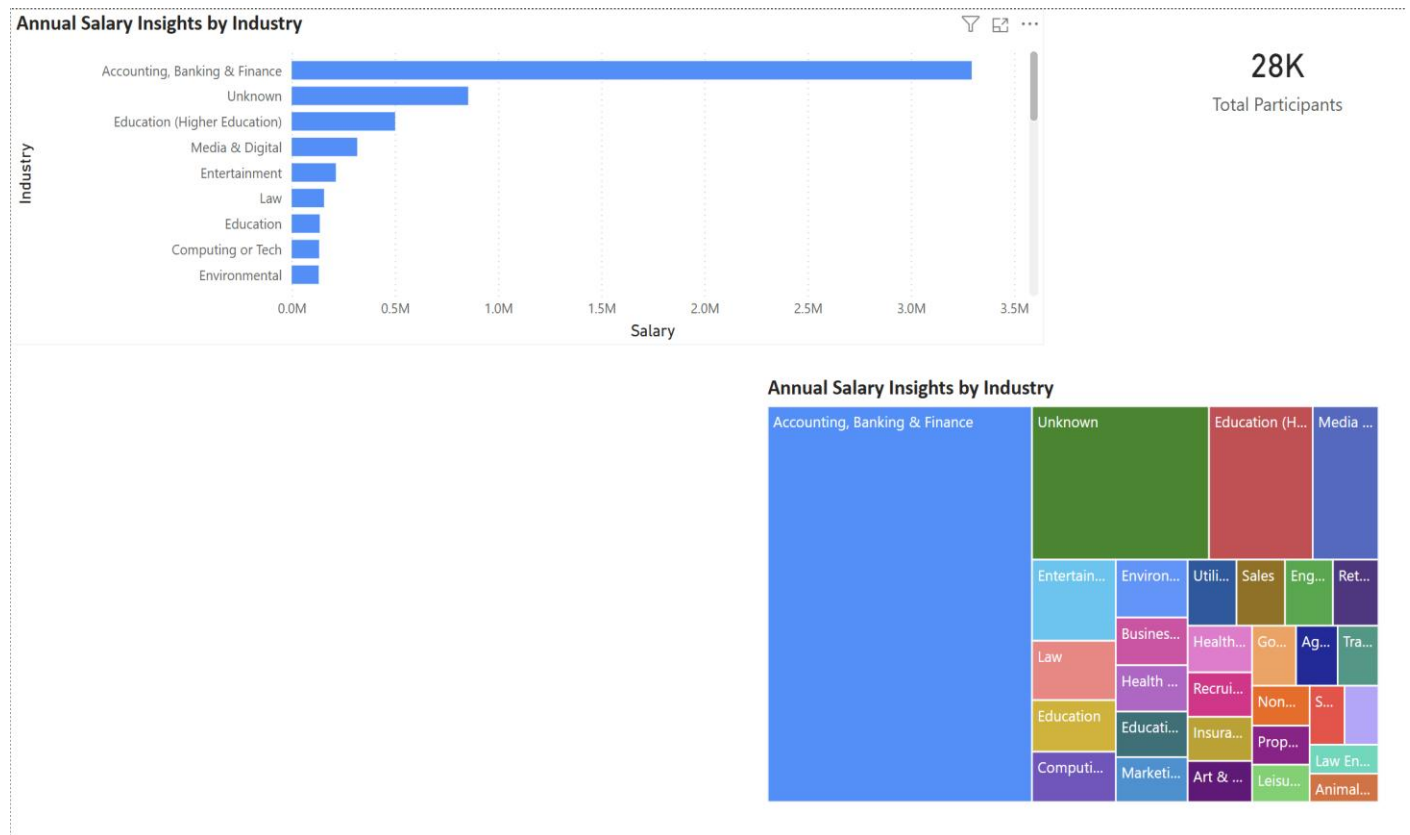
Industry Gender Breakdown							
	Man	Non-binary	Other or prefer not to answer	Prefer not to answer	Unknown	Woman	Grand Total
Accounting, Banking & Finance	338	25	25		13	1473	1874
Agriculture or Forestry	18	10	3	1		111	143
Animal Welfare	3	1				31	35
Art & Design	64	17	4		1	279	365
Business or Consulting	177	29	11		9	818	1044
Computing or Tech	2102	164	60		27	2814	5167
Education	48	27	6		6	448	535
Education (Higher Education)	304	71	30		19	2047	2471
Education (Primary/Secondary)	73	20			6	737	836
Engineering or Manufacturing	503	33	24		16	1318	1894
Entertainment	59	10	8			179	256
Environmental	21	2			1	57	81
Government and Public Administration	265	49	22		6	1679	2021
Health care	197	39	22		13	1632	1903
Healthcare	31	2	1		2	123	159
Hospitality & Events	53	18	3		2	243	319
Insurance	91	10	4		4	424	533
Law	117	17	10		11	951	1106
Law Enforcement & Security	10	2				31	43
Leisure, Sport & Tourism	16	1				88	105
Marketing, Advertising & PR	149	19	7		7	954	1136
Media & Digital	150	24	10		5	593	782
Nonprofits	214	86	21		13	2094	2428
Property or Construction	89	6	4		3	371	473
Recruitment or HR	33	7	2		4	437	483
Retail	100	24	11		4	407	546
Sales	78	8	1		2	237	326
Social Work	11	10	1		1	254	277
Transport or Logistics	79	8	4			218	309
Unknown	27		1		3	124	155
Utilities & Telecommunications	102	7	1		1	245	356
Grand Total	5522	746	296	1	179	21417	28161

Experience Level & Age Category:

- Advanced Professionals form the largest group (9,639), concentrated mainly in the Adult (6,997) and Young-Adult (1,904) categories.
- Early Career professionals (3,054) are strongly represented in Young-Adult (2,176) and Youth (764) groups, reflecting entry-level demographics.
- Experienced individuals (5,385) are mostly Young-Adults (4,094), showing a strong mid-career presence in younger age ranges.
- Experts are a smaller group (125) but heavily represented in Senior (81) and Elder (40) categories, consistent with advanced specialization.

Experience Level & Age Category Distribution								
	Adult	Elder	Junior	Mid-Senior	Senior	Young-Adult	Youth	Grand Total
Advanced Professional	6997	5		669	61	1904	3	9639
Beginner	16		3	4	1	188	337	549
Early Career	83		6	19	6	2176	764	3054
Experienced	1198	1		61	11	4094	18	5383
Expert	1	40	2		81	1		125
Intermediate	404	1	1	27	7	4318	137	4895
Senior Professional	1208	13	2	2109	306	6	1	3645
Veteran	4	35	1	304	524	2	1	871
Grand Total	9911	95	15	3193	997	12689	1261	28161

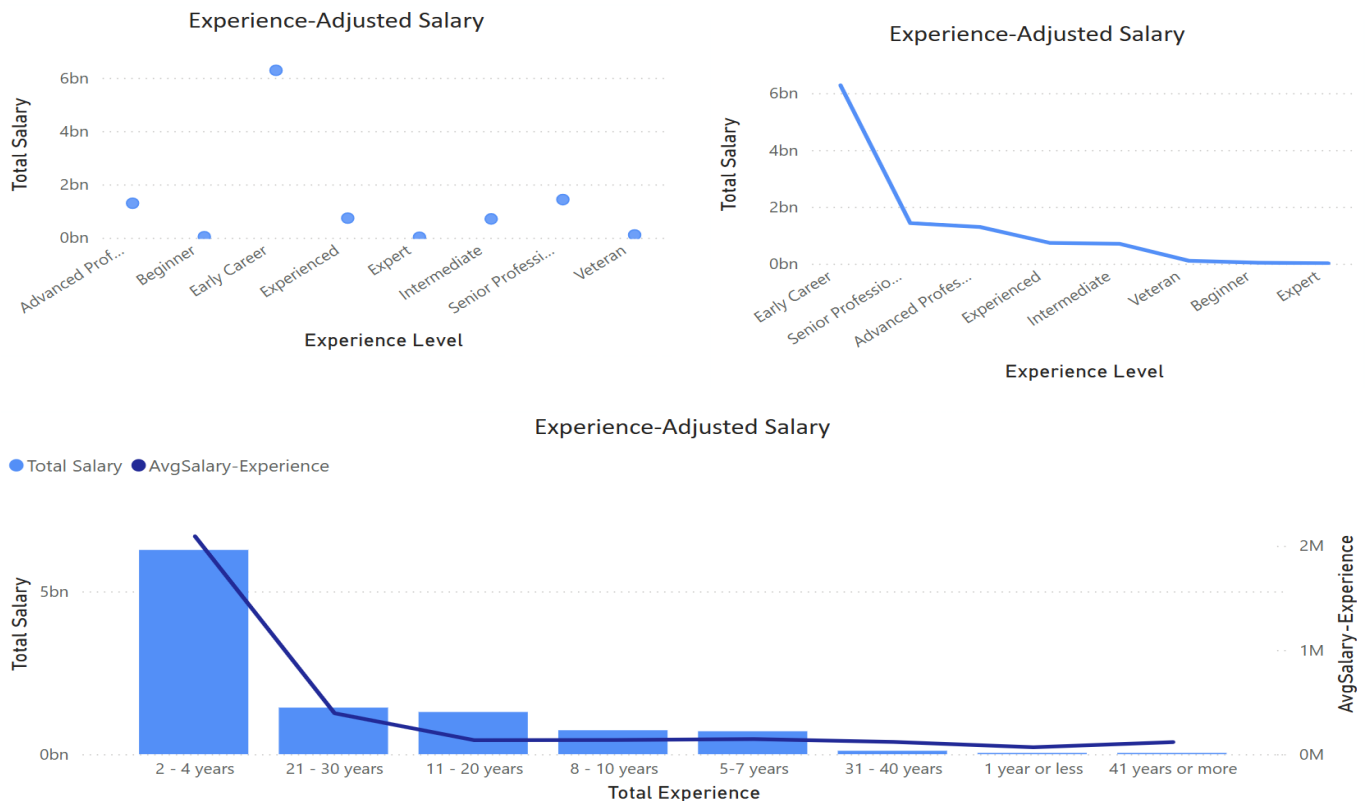
Visualization:



Annual Salary Insights by Industry:

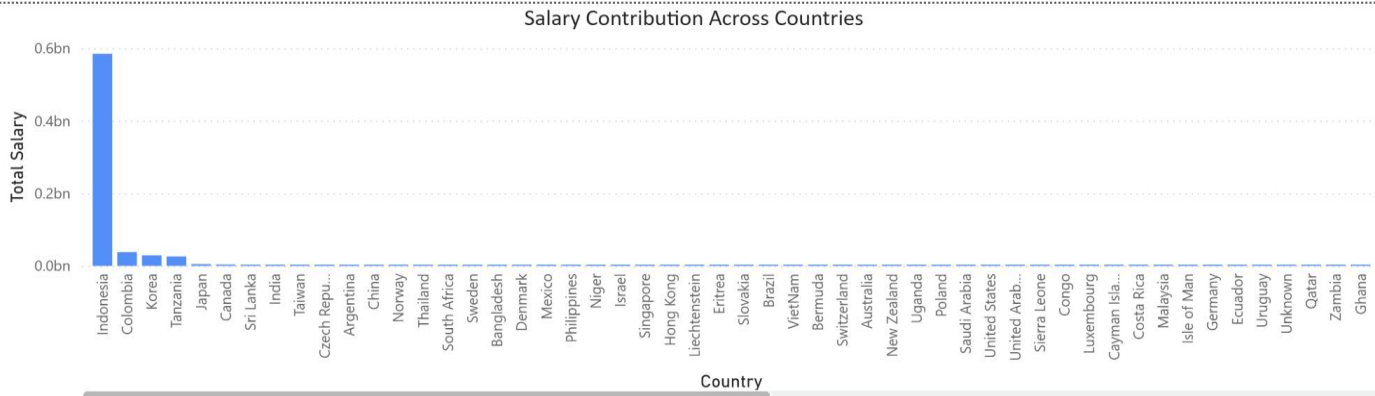
- Accounting, Banking & Finance shows the highest average annual salary among all industries.
- Unknown entries rank second, indicating some unclassified records with high salary values.
- Education (Higher Education) also reports strong average salaries compared to other sectors.
- Industries like Media & Digital and Entertainment fall in the mid-range.
- Computing/Tech has competitive salaries but not the highest compared to finance.
- Overall, salary distribution varies widely, with finance leading and service-oriented industries showing lower averages.

Experience-Adjusted Salary:

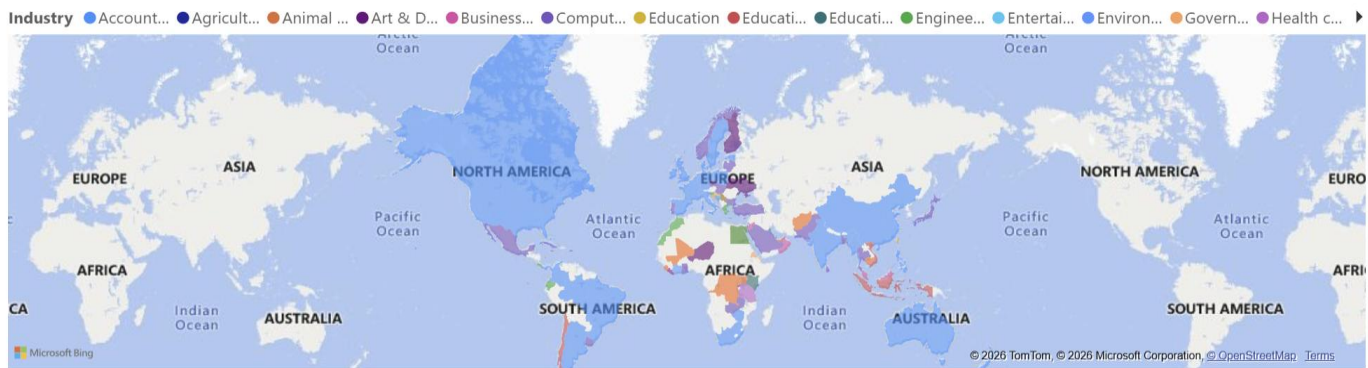


- Participants with 2–4 years of experience contribute the highest total salary, showing strong early-career earning potential.
- The 21–30 year and 11–20 year ranges also generate significant salary totals, reflecting mid-career stability.
- The average salary per year of experience is highest in the 2–4 year range, then drops sharply, suggesting early efficiency in earnings.
- Beyond 10 years of experience, average salary efficiency remains relatively low, indicating plateauing compensation despite longer tenure.
- Very long experience ranges (31–40 years, 41+ years) show modest total salary contributions and low efficiency, highlighting diminishing returns over extended careers.

Salary Contribution Across Countries:



Country-level Industry



Salary Contribution Across Countries:

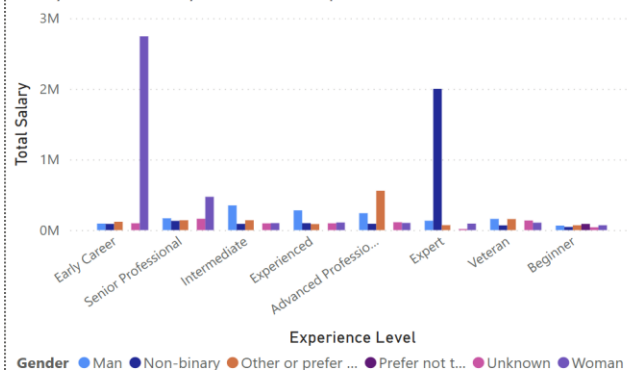
- Indonesia contributes the largest share of total salary (~0.6bn), far surpassing all other countries.
- Other notable contributors include Colombia, Korea, Tanzania, Japan, and Canada, though their totals are much smaller compared to Indonesia.
- Most remaining countries show relatively modest salary contributions, highlighting a strong geographic concentration of salary data.

Country-level Industry:

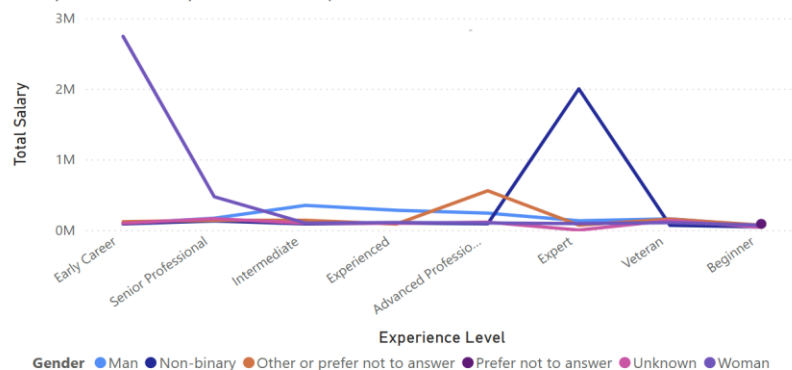
- Each country shows a dominant industry, reflecting local specialization and workforce distribution.
- Accounting, Banking & Finance appears as a leading industry in several countries, aligning with its high salary averages.
- Education (Higher Education) and Business/Consulting are also prominent in multiple regions, indicating global demand for these sectors.
- The map highlights diversity of industry dominance worldwide, showing how economic focus varies by region.

Salary Distribution by Gender and Experience:

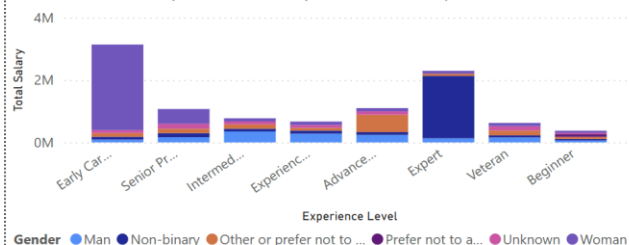
Salary Distribution by Gender and Experience



Salary Distribution by Gender and Experience



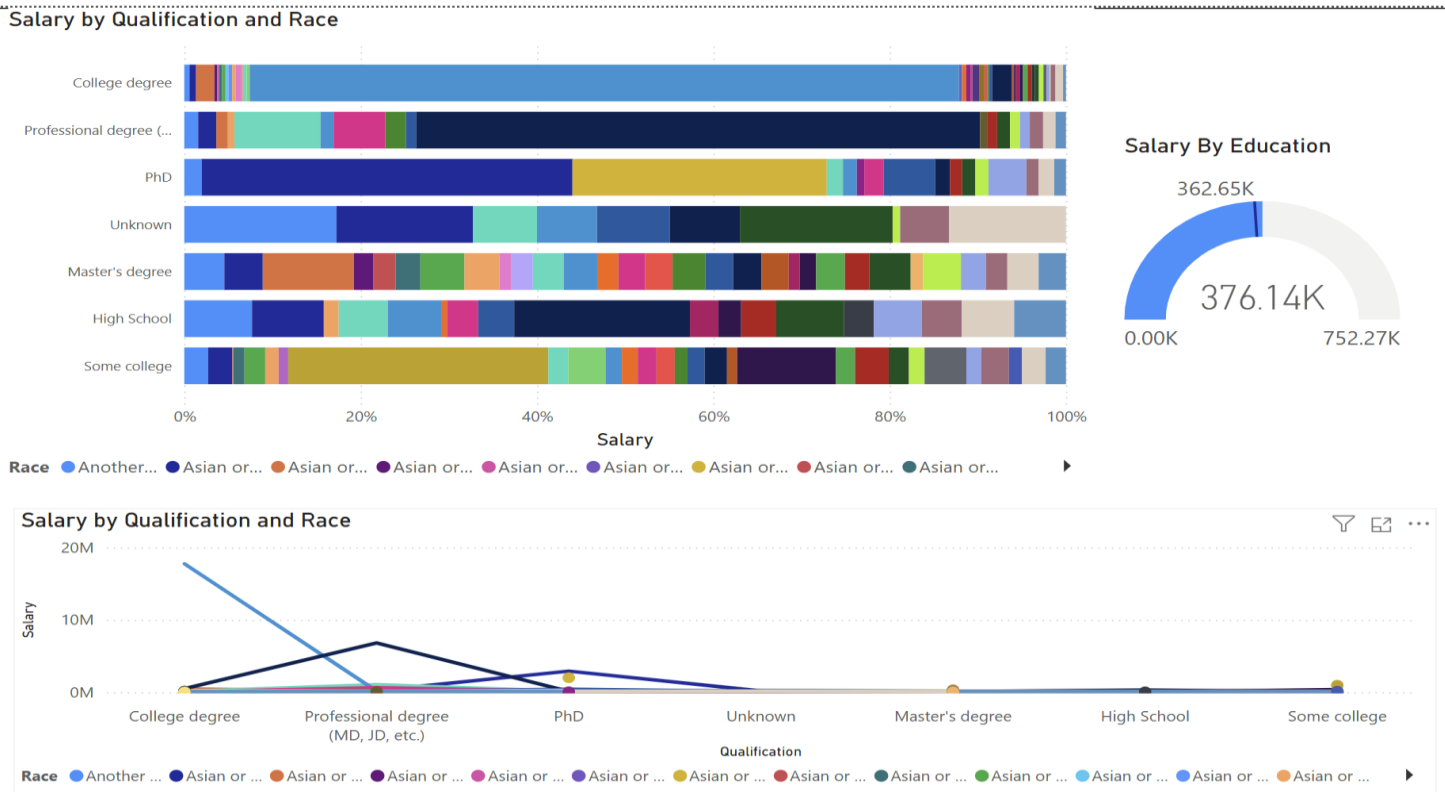
Salary Distribution by Gender and Experience



Experience Level	Man	Non-binary	Other or prefer not to answer	Prefer not to answer	Unknown
Veteran	158916.32	67333.33		156915.38	136607.14
Senior Professional	167424.61	130973.61		139729.27	161032.20
Intermediate	349857.64	88556.45		140245.73	96002.33
Expert	133180.56	200000.00		70366.67	5.00
Experienced	279799.00	98607.97		85968.98	96484.38
Early Career	92294.18	88121.77		118051.27	97028.96
Beginner	63933.42	44665.55		68955.56	88010.00
Advanced Professional	239647.25	89691.59		557468.50	111618.96
Total	229221.83	94583.51		246691.01	88010.00

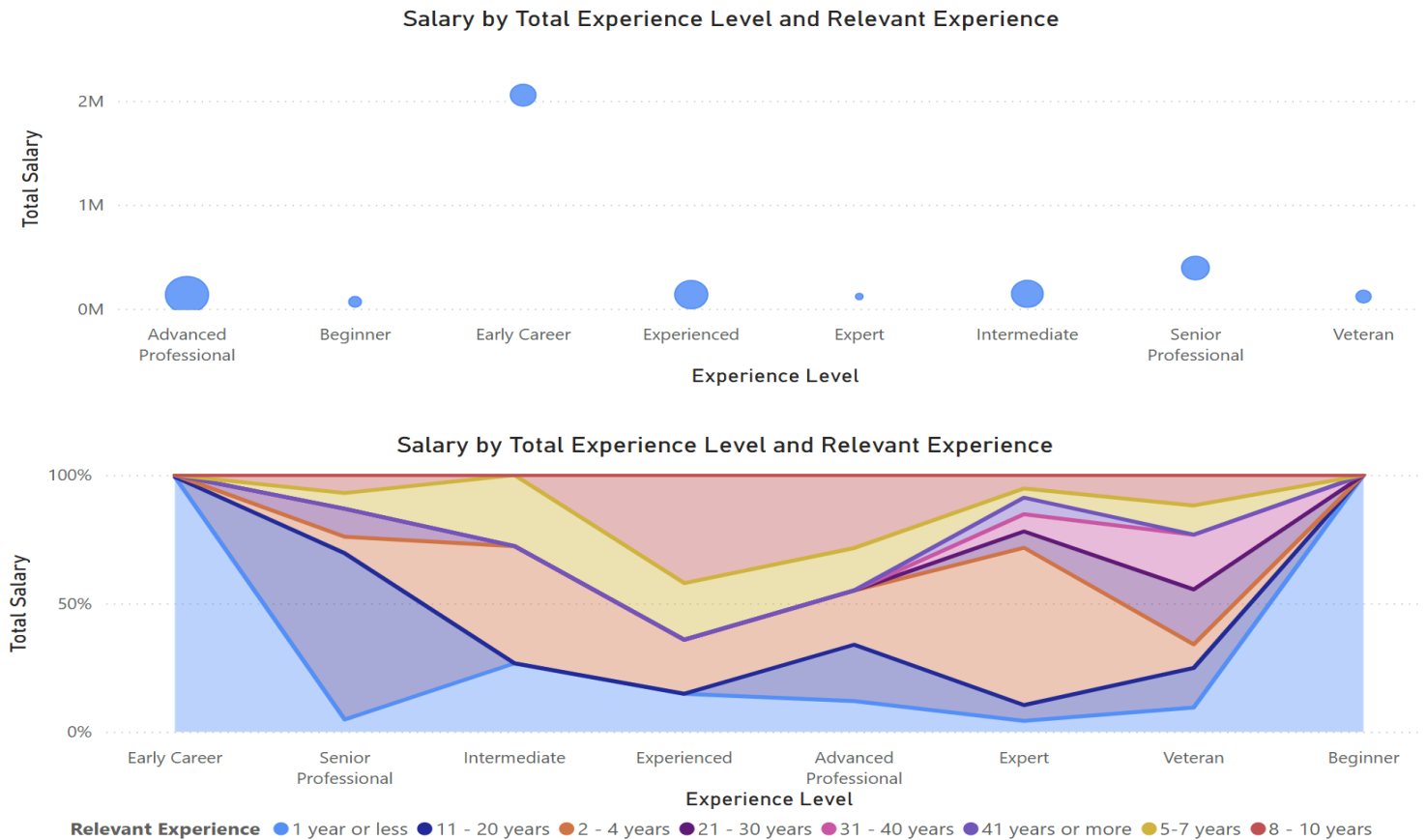
- Advanced Professionals show the highest average salary for “Other or prefer not to answer” (557,468.50), which is significantly above all other gender categories.
- Intermediate and Experienced levels contribute large salary totals for men (349,857.64 and 279,799.00 respectively), reflecting strong mid-career compensation.
- Non-binary salaries peak at the Expert level (200,000.00), though totals are smaller compared to other categories.
- Women are not included in this table, but the “Unknown” category shows notable spikes, especially in Senior Professional (161,032.20) and Veteran (136,607.14) levels.
- Beginner salaries are the lowest overall, with men at 63,933.42 and “Unknown” at 40,298.44, showing limited compensation at entry level.
- Overall, salary distribution varies widely by both gender and experience level, with certain categories (like “Other” and “Unknown”) showing unusually high averages that may indicate data anomalies or niche roles.

Salary by Qualification and Race:



- Master's degree holders show strong salary averages across multiple racial categories, reflecting the premium value of postgraduate education.
- PhD and Professional degree (MD, JD, etc.) groups have high average salaries but smaller participant counts, meaning their overall contribution to total salary is limited compared to larger groups like College or Master's.
- High School and Some College qualifications contribute lower salary averages, consistent with limited earning potential at lower education levels.
- The stacked bar chart shows racial disparities within each qualification: some race categories consistently earn higher averages, while others cluster at lower ranges.
- The line chart trend highlights that salary generally increases with higher qualifications, but the slope varies by race, suggesting unequal returns on education.
- The gauge chart indicates an overall average salary of 376.14K, slightly above the benchmark value of 362.65K, showing that the dataset's average is performing better than the set target.

Salary by Total Experience Level and Relevant Experience:

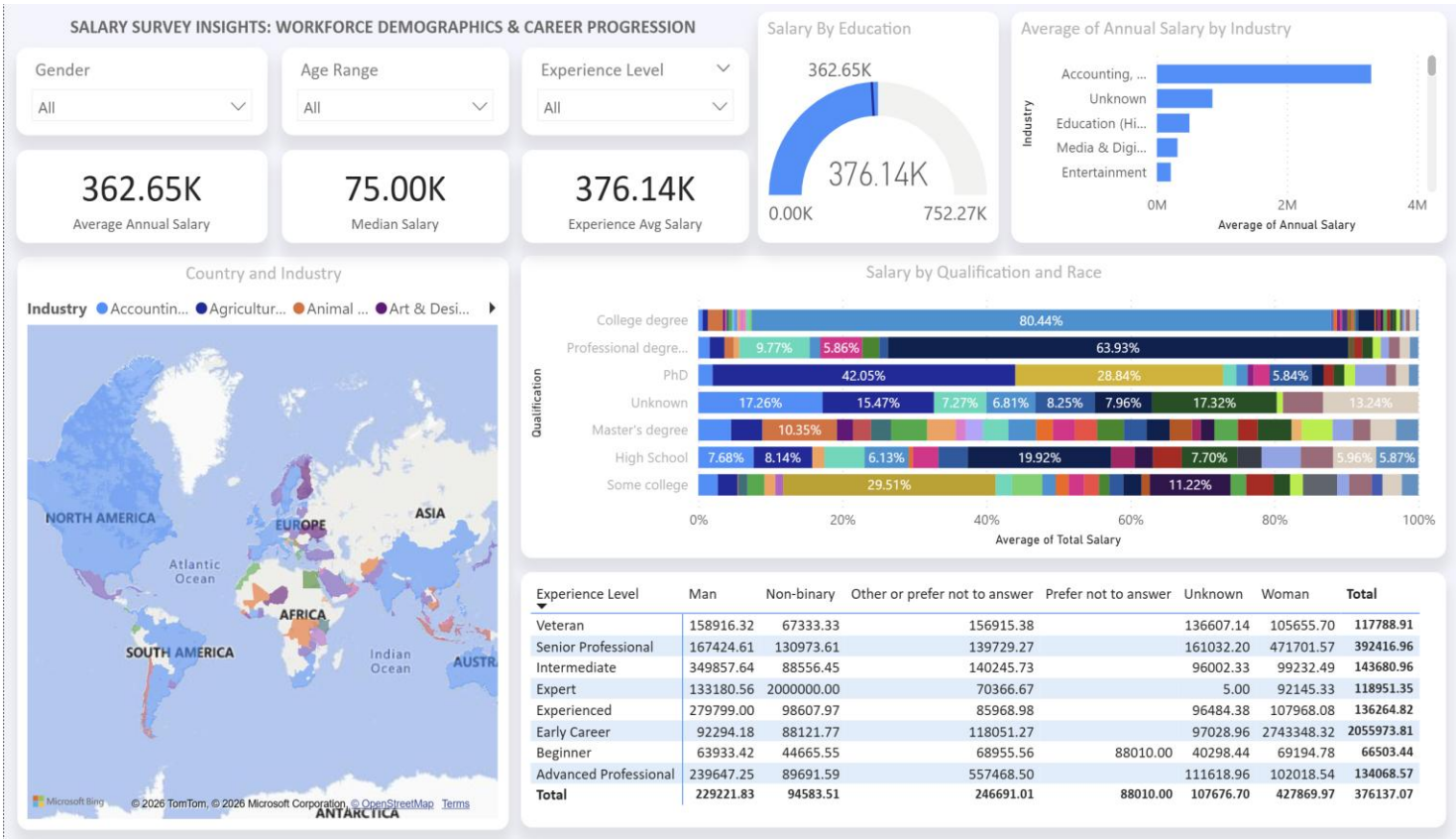


Bubble Chart – Salary by Experience Level

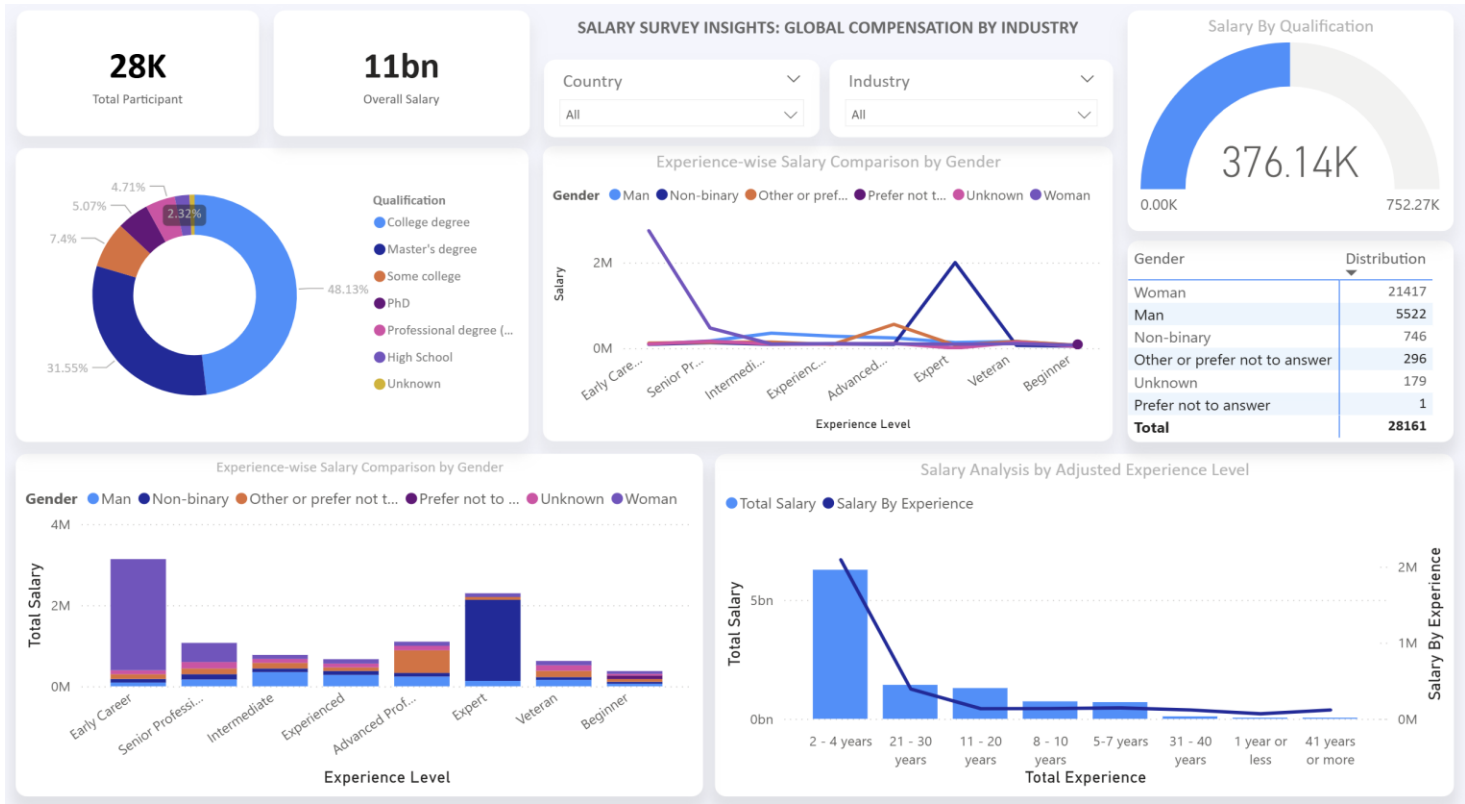
- Advanced Professionals and Senior Professionals show the largest bubbles, meaning they contribute the highest total salary overall.
- Intermediate and Experienced levels also contribute significantly, reflecting strong mid-career compensation.
- Beginner and Early Career levels have smaller bubbles, consistent with lower salary totals at entry stages.
- Experts and Veterans show moderate contributions, likely due to smaller participant counts but higher individual salaries.

Dashboard:

Workforce salaries vary significantly by qualification and experience, with clear gender distribution patterns influencing career progression.



Experience level plays a critical role in salary progression, with advanced professionals and veterans consistently earning more than beginners and early-career employees.



Conclusion:

The Salary Survey Insights mini project successfully cleaned and transformed raw survey data to ensure accuracy and reliability. Interactive dashboards were created in Power BI to visualize salary trends across demographics, qualifications, industries, and work locations. The insights highlight how education, experience, and industry choices influence compensation, while also reflecting workforce diversity. Overall, the project meets its objective of turning survey data into clear, actionable insights for understanding global salary patterns.